Toronto Metropolitan University

Department of Electrical, Computer and Biomedical Engineering

COE768

Midterm Examination **Oct. 20, 2025**

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Time: 1.5 hours

Instructions

This exam paper contains 8 questions. Please check your copy to make sure that it contains 10 pages (including this page) before you start.

(1) Closed book
(2) No cellphone or Internet connected devices.
(3) No questions during the exam. If in doubt on any question, then you must clearly state your own assumption(s).
(4) For each question, you must show your steps and reasoning clearly to be awarded part marks for your answer. You must provide formulas if they are used in each step of your answers.

For marking use only (do not fill)

Questions	Score	Max
1	11	12
2	4	18
3	2	10
4	5	10
5	2	10
6	2	10
7	~~8~~ 7	15
8	4	15

Total 100 37

Formulas you may need:

- Transmission latency T_f: M/R (M: data size in bits and R: bit rate)
- Propagation latency T_c: the ratio between link length and the propagation speed
- Link efficiency of stop and wait protocol: $1/(1+2a)$ where $a = T_c/T_f$
- Link efficiency of CSMA/CD: $1/(1+5a)$ where $a = T_c/T_f$

Metric Units
1 K(ilo) bit = 10^3 bits
1 M(ega) bit = 10^6 bits
1 G(iga) bit = 10^9 bits

Kbps: Kilobits per second
Mbps: Megabits per second

Second (s), Millisecond (ms) and Microsecond (μs)

$1s = 10^3$ ms
$1s = 10^6$ μs

Q1. For each of the following questions, circle T (true) or F (false). (12 marks)

T F Stop and wait protocol can be implemented in the application layer.
T F Bit stuffing may expand the size of the frame.
T F In the OSI Reference Model, a protocol defines the mechanism for vertical communication between adjacent layers.
T F In CSMA/CD, the amount of time that it takes to detect a collision could be greater than the propagation delay.
T F The longer the frames or the longer the propagation time, the higher link efficiency can be achieved in CSMA/CD.
T F The bandwidth delay product of a link is how long it takes to send a frame to the other side and get a response back.
T F Switches only forward packets that are addressed to them.
T F Switches decrement the TTL field in the IP header.
T F The multiple access protocol is a function of the data-link layer.
T F UDP provides connectionless service and delivers packets quickly. In case of packet loss, UDP does not provide retransmission.
T F UDP is preferred over TCP for transferring a real-time voice over IP networks.
T F The application-layer protocol for the World Wide Web is HTML.

Q2. Multiple choices (Select only one answer) (18 marks)
(1) "Layering" is commonly used in computer networks because:
a. Encapsulation is the lowest overhead method to transmit data.
b. It allows functionality re-use.
c. It enables network to run faster.
d. All of above

(2) If a network uses CSMA/CD, we require transmission delay Tf >2Tc (propagation delay) because:

a. Tc is the round-trip time from a source to the destination and back again. Therefore, it must be at least twice the one way propagation delay.

b. Otherwise, the signal would degrade too much along the wire making it difficult to detect collisions.

c. In any network (regardless of whether we use CSMA/CD or not) the transmission time of a frame is a function of both the bit rate, and speed of propagation along the wire.

d. none of above

(3) When a HTTP server is not communicating with any web browsers (clients), how many sockets are created on the server:

a. 0

b. 1

c. 2

d. 10

(4) In a UDP-based server, some typical system calls (Socket API) used are:

a. send() and recv() (in that order)

b. recv() and send() (in that order)

c. sendto() and recvfrom() (in that order)

d. recvfrom() and sendto() (in that order)

(5) The Application layer provides services to which layer?

a. Transport layer

b. Data Link Layer

c. Network layer

d. none of the above

(6) The format of the messages exchanged between physical devices on a network are defined by:

a. the transport layer

b. end systems

c. a protocol

d. sockets

e. an Internet Service Provider (ISP)

(7) The main function of the network layer is _____.

a. to control the error rate between the transmitter and receiver

b. to secure the data between the source and the destination

c. to find the best path from the source to the destination

d. to manage the session between the source and the destination

(8) After the kth consecutive collision, each colliding station waits for a random time chosen from the interval-

a. (0 to 2^k) * 2Tc

b. (0 to 2^k-1) * 2Tc

c. (0 to 2^k-1) * Maximum Propagation delay

d. (0 to 2^{k-1}) * Maximum Propagation delay

(9) The Cyclic Redundancy Check (CRC) used on frames of Ethernet provides:
a. very powerful error detection
b. very powerful error correction
c. very powerful error detection and error correction
d. none of them

(10) Routers and switches are different because:
a. routers operate at the network layer, while switches operate at the data link layer
b. switches operate at the network layer, while routers operate at the data link layer
c. routers are for wired networks, while switches are for wireless networks
d. switches are for wired networks, while routers are for wireless networks

(11) In a TCP-based server, some typical system calls (socket API) used are:
a. socket() and connect() (in that order)
b. connect() and socket() (in that order)
c. accept() and listen() (in that order)
d. listen() and accept() (in that order)
e. recvfrom() and sendto() (in that order)

(12) Ethernet uses _____ MAC addresses that are imprinted on computers' network interface cards (NICs).
a. 32-bit
b. 64-bit
c. 6-byte
d. none of the above

12 (8) =

(13) What is the transmission delay of a packet with 8 kilobits on a 2 Mbps link?
a. 0.40 seconds
b. 4000 milliseconds
c. 0.04 seconds
d. none of the above

(14) When a collision is detected in a network using CSMA/CD,
a. The frame is immediately resent
b. A jam signal is sent by the station
c. The collision number is set to 0
d. The collision number is decremented by 1

(15) The protocol data unit(PDU) for the application layer in the Internet reference model is
a. Segment
b. Datagram
c. Message
d. Frame

(16) Which of the following transport layer protocols is used to support electronic mail?
a. HTTP
b. IP
c. TCP
d. UDP

(17) Which of the following is true about encapsulation

a. Packets encapsulate frames.

b. Messages encapsulate datagrams.

c. Frames encapsulate segments.

d. None of the above.

(18) What does traceroute measure?

a. end-to-end delay

b. transmission delay

c. round trip time

d. propagation delay

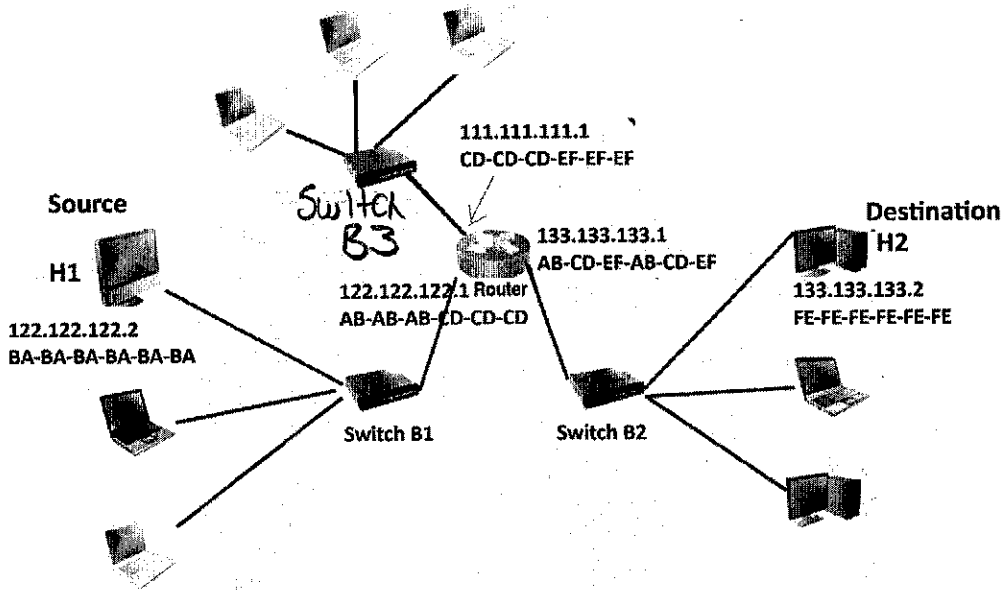
2 Q3. Describe two types of frame errors that occur in the transmission of frames. (10 marks)

Sequential errors

time-bound errors

Q4. (10 marks)

Following is a picture of a network with 3 switches and 1 router. Each interface of the router is labeled with an IP address and a MAC address. IP and MAC addresses of the hosts H1 and H2 are also given. Imagine that host H1 is sending a packet to host H2. Please answer the following questions about this figure:



How many local area networks are shown above?

Each personal device is a LAN.

∴ there are nine (9) LAN

Just before the packet reaches switch B1, what is its destination MAC address?

AB-AB-AB-CD-CD-CD

The destination MAC address is given above, before the packet reaches Switch B1.

Just before the packet reaches switch B2, what is its source MAC address?

AB-CD-EF-AB-CD-EF

The Source MAC address is given above, before the packet reaches Switch B2.

Just after the packet leaves the router. what is its source IP address?

133.133.133.2

The Source IP address after the packet leaves the router is given as:

133.133.133.2

Q5. (10 marks) Transport (2)

At which layers of the Internet reference model are CRC and checksum performed?

(2) The data link layer (layer 2)

+1

We already have CRC for error detection, why still need Checksums in the layers of Internet reference model?

The CRC error detection method is the cyclic redundancy check. This method is referenced by Question 6

- zero (error)
- non-zero (error)

+1

The checksum uses the sum of

- sender
  - * m bit packets of message (sums of packets
  - * m bit packet of sum + k addition bits
  - * 1's complement provides checksum

sum →

- Receiver
  - * sum + the checksum
  - * if 1's comp
    - zero (no errors)
    - non-zero (error)

(2) Q6. (10 marks)

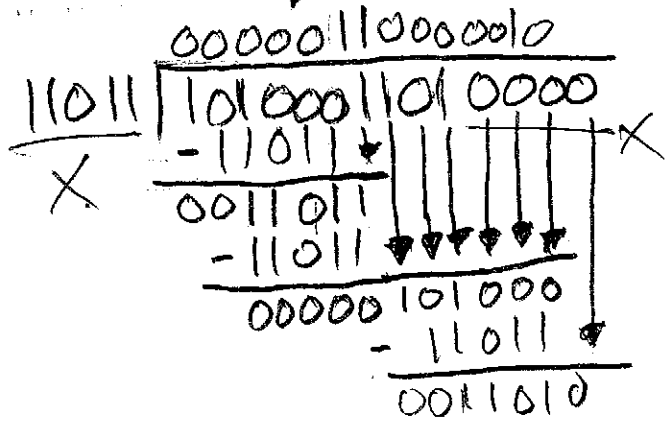
(a) Given a message M = 1010001101, determine the CRC using the generator polynomial $x^5 + x^4 + x^2 + 1$. Show your work.

CRC Generator = 11011

*Since $x^5 + x^4 + x^3 + x^2 + 1$

• Using the Cyclic Redundancy Check (CRC)

*NOTE that the CRC generator has 5 terms, thus, there will be an extra 4 bit sequence



∴ the CRC (Cyclic Redundancy Check) is equal to the remainder of the binary division.

CRC = 11010

(b) What is the transmitted message?

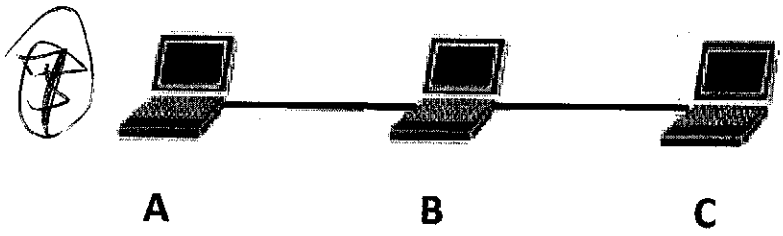
The transmitted message is equal to the binary sum of the CRC and the resultant of the binary division

$$\begin{array}{r} 11000010 \\ +\ 11010 \\ \hline 11001\,1100 \end{array}$$

∴ the transmitted message is 11001 1100.

Q7. (15 marls)

Suppose we have M bytes to send over a 3-node network as shown below.



$T_c = T$
$H(\text{header}) = N$
R bps

Assume that there is no processing delay and no queuing delay at all. The propagation delay is T seconds for both link A-B and link B-C. The transmission rate is R bps for both link A-B and link B-C. Suppose the amount of header required to add to each packet in order to transmit the data over the network is N bytes. Show steps in details. Provide formulas if they are used in your calculations.

(1) How much time does it take to send all M bytes of data as one packet from node A to node C and completely received by C?

*NOTE: 1 byte = 8 bits

Transmission Delay: $T_f = \frac{M+N}{R}$

∴ the time is given by the transmission delay

$T_f = \frac{M+N}{R} + 1 + 1$

(2) Divide the M bytes into two equal parts. After sending the first part, A immediately sends the second part. How much time does it take to send all the data as two packets to C (From the start of sending first packet at A till the second packet is completely received by C)?

Using the Stop and wait protocol, $\frac{1}{1+2a}$ where $a = \frac{T_c}{T_f}$

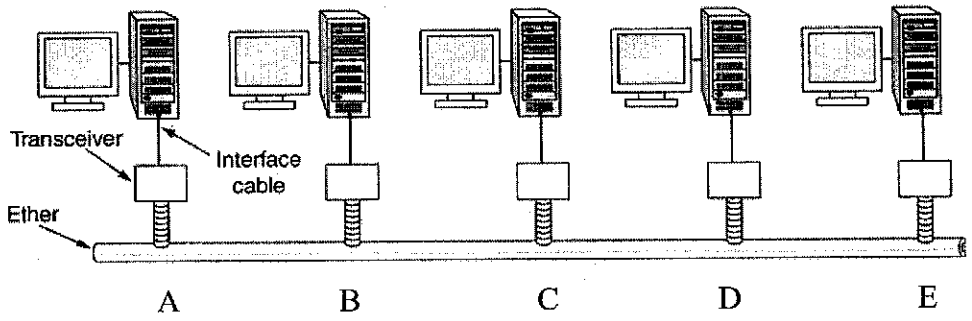
- Process 1
  - *no delay time
- Process 2 +2
  - *delay of T_f for process 1
  - *Transmission of process 2

$T_f = 2\left(\frac{\frac{M}{2}+N}{R}\right) + 3 = \frac{M+2N}{R}$ ∴ $T_f = \frac{M+2N}{R}$

Q8. (15 marks)

In a CSMA/CD network with a data rate of 10 Mbps (shown below), the maximum distance between any station pair (such as AB, AC, DE, etc) is found to be 2500m in order to make the collision detection work. What the maximum distance between any station pair should be if we increase the data rate to 100 Mbps? To 1 Gbps?

We assume that the frame size does not change and signals travel over the link is at a speed of $2x10^8$ meters per second. Show steps in details. Provide formulas if they are used in your calculations.



Given: bit rate(R) = 10 Mbps & 100 Mbps & 1 Gbps
Length(l) = 2500m
Propagation speed (c) = $2x10^8$ m/s

Required:
- Maximum distance?

Analysis:
- Transmission latency ↳ $T_f = \frac{M}{R}$
- Propagation latency ↳ $T_c = \frac{l}{c}$
- Link efficiency of CSMA/CD ↳ $\frac{1}{(1+5a)}$ *where ↳ $a = \frac{T_c}{T_f}$

Statement:
2500(0.167)
= 416.67 m
∴ max distance is 416.67 m

+4

Solution:

$T_c = \frac{2500}{2x10^8} = 1.25x10^{-5}$ s

$T_{f_1} = \frac{125}{10M} = 1.25x10^{-5}$ s, $T_{f_2} = \frac{125}{100M} = 1.25x10^{-5}$ s, $T_{f_3} = \frac{12500}{1G} = 1.25x$ X

∴ $T_f = 1.25x10^{-5}$ s

$a = \frac{1.25x10^{-5}}{1.25x10^{-5}} = 1$

CSMA/CD = $\frac{1}{1+5a} = \frac{1}{6} = 0.167$

T_{f_3} | T_{f_2} | T_{f_1}

1 Gbps / $2x10^8$ = 5(2500) = 12500

100 Mbps / $2x10^8$ = 0.5(2500) = 1250

10 Mbps / $2x10^8$ = 0.05(2500) = 125